# Differential Drive Mobile Robot Simulation and Dead Reckoning

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**Abstract**

This report presents the implementation and validation of a differential-drive mobile robot simulator and a dead-reckoning localization method. Planar direct and inverse pose compounding were first implemented and incorporated into the simulation and localization modules. Lab 1 implements the planar motion model and the simulated encoder and compass measurements. Lab 2 converts the encoder readings into body motion and integrates them to estimate the robot pose through dead reckoning. Circular and figure-eight trajectories are used to compare different velocity gains and encoder-noise levels. The results show that the tested gains produce similar long-term paths, whereas increasing the encoder noise causes increasing localization drift.

## 1 Introduction

The objective of Labs 1 and 2 is to build the simulation and odometric localization foundations required by the later probabilistic localization exercises. The Pre-Lab covered the planar pose representation and the direct and inverse pose-compounding operators. Lab 1 then applied these operations to the differential-drive simulator, wheel-encoder model, and commanded trajectories. Lab 2 used the simulated encoder readings to estimate the robot pose through dead reckoning.

The implementation uses a world frame, denoted by $N$, and a body-fixed robot frame, denoted by $B$. A planar robot pose contains two Cartesian coordinates and one yaw angle. Lab 1 produces the simulated or ground-truth state, whereas Lab 2 reconstructs the pose using only incremental encoder information.

The report is organized accordingly. The remainder of this introduction describes the software structure and the Pre-Lab pose operations. Section 2 presents the differential-drive simulation, and section 3 describes the dead-reckoning implementation. The experimental comparison is discussed in section 4, followed by the conclusions in section 5.

### 1.1 Development environment and software structure

Jupyter Notebook was used only in Lab 0 to implement and validate the direct and inverse pose-compounding operations, $\oplus$ and $\ominus$. These operations were subsequently incorporated

into the Python modules used in Labs 1 and 2. NumPy provides vector and matrix operations, Matplotlib is used for visualization, and Robotics Toolbox supplies the animated vehicle representation.

The relevant software components and their roles are summarized in table 1.

Table 1: Main Python files used in Labs 1 and 2.

| File | Responsibility |
|---|---|
| `Pose3D.py` | Represents a planar pose and implements direct and inverse pose compounding. |
| `SimulatedRobot.py` | Defines the common simulation interface, state storage and visualization infrastructure. |
| `DifferentialDriveSimulatedRobot.py` | Implements the differential-drive motion model and simulated sensors. |
| `Localization.py` | Provides the iterative localization loop and trajectory visualization. |
| `DR_3DOFDifferentialDrive.py` | Converts encoder readings into body motion and performs dead-reckoning integration. |
| `main.py` | Configures the map and initial state, creates the simulated and dead-reckoning robots, and runs the localization sequence. |

## 1.2 Pre-Lab: Planar pose operations

### 1.2.1 Planar pose representation

A three-degree-of-freedom pose is represented by

$$\boldsymbol{\eta} = \begin{bmatrix} x & y & \psi \end{bmatrix}^T \in \mathbb{R}^3, \tag{1}$$

where $(x, y)$ is the position and $\psi$ is the yaw angle. Its equivalent homogeneous transformation is

$$T(\boldsymbol{\eta}) = \begin{bmatrix} \cos\psi & -\sin\psi & x \\ \sin\psi & \cos\psi & y \\ 0 & 0 & 1 \end{bmatrix}. \tag{2}$$

This matrix form explains the pose operations used throughout the simulator: direct compounding corresponds to multiplying transformations, and inverse compounding corresponds to inverting a transformation.

### 1.2.2 Direct pose compounding

Let ${}^A\boldsymbol{x}_B$ describe frame $B$ relative to frame $A$, and let ${}^B\boldsymbol{x}_C$ describe frame $C$ relative to $B$. The pose of $C$ relative to $A$ is

$${}^A\boldsymbol{x}_C = {}^A\boldsymbol{x}_B \oplus {}^B\boldsymbol{x}_C, \tag{3}$$

which is equivalent to $T({}^A\boldsymbol{x}_C) = T({}^A\boldsymbol{x}_B)T({}^B\boldsymbol{x}_C)$. In coordinate form,

$$\begin{bmatrix} {}^Ax_C \\ {}^Ay_C \\ {}^A\psi_C \end{bmatrix} = \begin{bmatrix} {}^Ax_B + {}^Bx_C\cos({}^A\psi_B) - {}^By_C\sin({}^A\psi_B) \\ {}^Ay_B + {}^Bx_C\sin({}^A\psi_B) + {}^By_C\cos({}^A\psi_B) \\ {}^A\psi_B + {}^B\psi_C \end{bmatrix}. \tag{4}$$

The $\oplus$ operator is used to compose poses or motions expressed in different reference frames. In this case, ${}^B\boldsymbol{x}_C$ cannot be added directly to ${}^A\boldsymbol{x}_B$ because its components are written in frame $B$. The operation first transforms them into frame $A$ and then combines both poses to obtain ${}^A\boldsymbol{x}_C$.

The implementation extracts the pose components, evaluates equation (4), and returns another `Pose3D` object. The essential operation is:

Listing 1: Core of the direct compounding implementation.

```
x_result = xB + xC * cos(theta_B) - yC * sin(theta_B)
y_result = yB + xC * sin(theta_B) + yC * cos(theta_B)
theta_result = theta_B + theta_C
return Pose3D(np.array([[x_result], [y_result], [theta_result]]))
```

### 1.2.3 Inverse pose compounding

Inverse compounding expresses frame $A$ relative to frame $B$ from the pose of $B$ relative to $A$:

$${}^B\boldsymbol{x}_A = \ominus{}^A\boldsymbol{x}_B = \begin{bmatrix} -{}^Ax_B\cos({}^A\psi_B) - {}^Ay_B\sin({}^A\psi_B) \\ {}^Ax_B\sin({}^A\psi_B) - {}^Ay_B\cos({}^A\psi_B) \\ -{}^A\psi_B \end{bmatrix}. \tag{5}$$

This is the coordinate form of $T({}^B\boldsymbol{x}_A) = T({}^A\boldsymbol{x}_B)^{-1}$.

The implementation evaluates equation (5) directly and returns the inverse pose as another `Pose3D` object:

Listing 2: Core of the inverse compounding implementation.

```
x_result = -xB * cos(theta_B) - yB * sin(theta_B)
y_result =  xB * sin(theta_B) - yB * cos(theta_B)
theta_result = -theta_B
return Pose3D(np.array([[x_result], [y_result], [theta_result]]))
```

After implementing and testing both operators in the Pre-Lab notebook, the same code was included in `Pose3D.py`. This makes $\oplus$ and $\ominus$ available as `Pose3D` methods for the differential-drive simulation developed in Lab 1.

# 2 Lab 1: Simulation

Lab 1 consists of completing the `DifferentialDriveSimulatedRobot` class, which inherits from `SimulatedRobot`. The base class provides the common simulation interface, state storage, and visualization tools, whereas the derived class implements the differential-drive motion and sensor models.

## 2.1 `__init__`: robot configuration

The constructor first calls `super().__init__` to initialise the state, map, sampling period, and visualization variables provided by `SimulatedRobot`. The simulated state contains the world-frame pose and body-frame velocity:

$$\boldsymbol{x}_{s,k} = \begin{bmatrix} \boldsymbol{\eta}_{s,k} \\ \boldsymbol{\nu}_{s,k} \end{bmatrix} = \begin{bmatrix} x_k & y_k & \psi_k & u_k & v_k & r_k \end{bmatrix}^T. \tag{6}$$

It then defines the parameters required by the differential-drive model. The wheel separation is $b = 0.5$ m, the wheel radius is $R = 0.1$ m, and the simulator initially assigns $N_p = 1024$ encoder pulses per wheel revolution. However, `main.py` creates the dead-reckoning object before starting the localization loop. Its constructor changes `robot.pulse_x_wheelTurns` to 4096 on the same shared robot object. Therefore, the effective resolution throughout the executed simulation is $N_p = 4096$ pulses per wheel revolution; 1024 is only the simulator's initial default value. The motion and encoder noise are described by

$$Q_s = \text{diag}\left(0.1^2,\, 0.01^2,\, \text{deg2rad}(1)^2\right), \qquad R_e = \text{diag}(22^2, 22^2). \tag{7}$$

Here, $Q_s$ is the covariance of the acceleration perturbation applied by the motion model, while $R_e$ is the covariance of the encoder noise in squared pulses. The remaining constructor parameters configure sensors that are not required for the simulation and dead-reckoning tasks discussed here.

## 2.2 `fs`: motion model

This method computes the new simulated state from the previous state and the commanded forward and angular velocities. The desired input is $\boldsymbol{u}_{s,k} = [u_d,\, r_d]^T$. Because a differential-drive robot cannot move laterally, the corresponding desired body velocity is $\boldsymbol{\nu}_d = [u_d,\, 0,\, r_d]^T$.

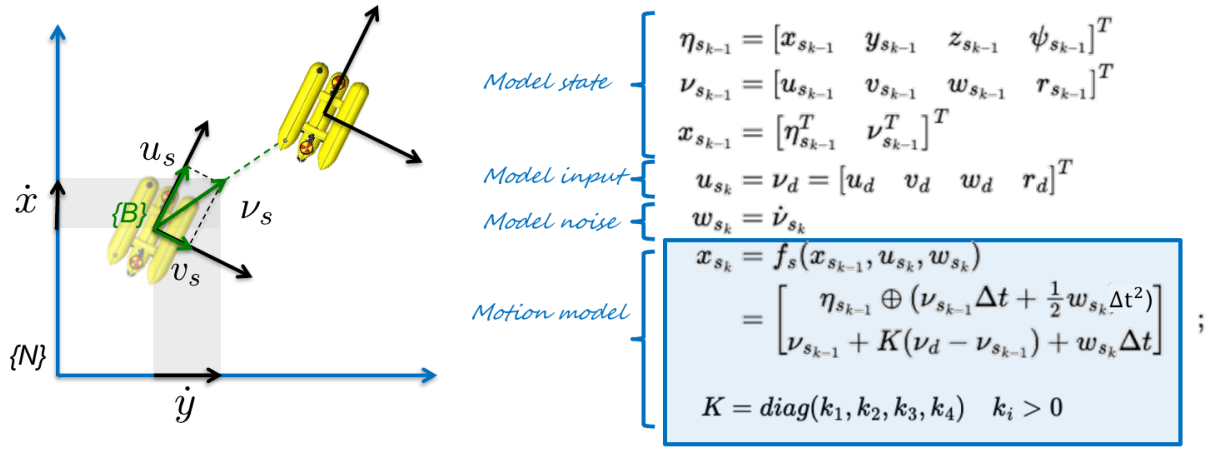



Figure 1: State, input, noise, and motion model used by `fs`.

The planar version of the model in figure 1 uses $\boldsymbol{\eta} = [x,\, y,\, \psi]^T$ and $\boldsymbol{\nu} = [u,\, v,\, r]^T$, with $v_d = 0$ because the wheels prevent lateral motion.

The nominal implementation uses $K = I_3$. Its effect on the velocity response is discussed in section 4.

The key lines in `fs` are therefore:

Listing 3: Pose and velocity update in the simulator.

```
w = np.random.multivariate_normal(np.zeros(3), self.Qsk).reshape(3, 1)
eta_new = eta_prev.oplus(Pose3D(
    nu_prev * self.dt + 0.5 * w * self.dt**2))
nu_new = nu_prev + K @ (nu_desired - nu_prev) + w * self.dt
```

## 2.3 `ReadEncoders`: wheel-encoder simulation

`ReadEncoders` converts the motion between two consecutive simulated poses into the measurements that would be produced by the left and right wheel encoders. The implemented sequence is summarized in figure 2.

**1. Relative motion in the previous body frame**

$$\Delta\boldsymbol{\eta}_k = (\ominus\boldsymbol{\eta}_{s,k-1}) \oplus \boldsymbol{\eta}_{s,k}$$

↓

**2. Extract the local forward displacement and yaw change**

$$d = \Delta x_k, \qquad \Delta\psi = \Delta\psi_k$$

↓

**3. Compute the distance traveled by each wheel**

$$d_L = d - \frac{b}{2}\Delta\psi, \qquad d_R = d + \frac{b}{2}\Delta\psi$$

↓

**4. Convert wheel distances into ideal encoder pulses**

$$\boldsymbol{z}_{\text{ideal}} = \frac{N_p}{2\pi R}\begin{bmatrix} d_L \\ d_R \end{bmatrix}$$

↓

**5. Add encoder noise and form the measurement**

$$\boldsymbol{\epsilon}_k \sim \mathcal{N}(\mathbf{0}, R_e), \qquad \boldsymbol{z}_{s,k} = \boldsymbol{z}_{\text{ideal}} + \boldsymbol{\epsilon}_k$$

Figure 2: Sequence implemented by `ReadEncoders`.

The use of $\ominus$ in the first step is essential. Both simulated poses are stored relative to the world frame, but an encoder measures the displacement in the robot frame. In transformation form, the required relative motion is

$$T(\Delta\boldsymbol{\eta}_k) = T(\boldsymbol{\eta}_{s,k-1})^{-1} T(\boldsymbol{\eta}_{s,k}). \tag{8}$$

The inverse operator removes the previous world-frame pose, and the following $\oplus$ expresses the current pose relative to that previous robot frame. A direct subtraction would not rotate the translational displacement into the body axes and would therefore give incorrect wheel motion when the robot is not aligned with the world frame.

For the executed experiment, $b = 0.5$ m, $R = 0.1$ m, and the effective encoder resolution is $N_p = 4096$ pulses per wheel revolution. Since `ReadEncoders` accesses the shared robot attribute after it has been updated, this is also the value used to generate the simulated pulses. The encoder covariance is $R_e = \mathrm{diag}(22^2, 22^2)$ pulses$^2$. The method returns $(\boldsymbol{z}_{s,k}, R_e)$, where $\boldsymbol{z}_{s,k} = [n_L, n_R]^T$ contains the noisy left and right encoder pulses and $R_e$ describes their measurement uncertainty.

## 2.4 Compass and robot visualization

`ReadCompass` adds zero-mean Gaussian noise to the true yaw and returns the resulting measurement together with its variance. `PlotRobot` updates the vehicle icon using the current simulated pose.

Listing 4: Essential compass and visualization operations.

```
yaw_true = self.xsk[2, 0]
yaw_measured = yaw_true + np.random.normal(0.0, self.v_yaw_std)
return yaw_measured, self.v_yaw_std**2

self.vehicleIcon.update([self.xsk[0], self.xsk[1], self.xsk[2]])
plt.pause(0.0000001)
```

## 2.5 Trajectory commands

The simulator is tested with a constant forward speed and opposite angular velocities. The first command produces circular motion; changing the angular velocity sign produces the second lobe of the requested figure-eight sequence. Each command is applied for approximately one complete revolution, $k_{\mathrm{steps}} = \mathrm{round}(2\pi/(0.03\Delta t)) = 2094$. The applied commands are $(u_d, r_d) = (0.5, 0.03)$ and $(0.5, -0.03)$. The second command starts from the final simulated state of the first, preserving continuity between the two trajectory segments.

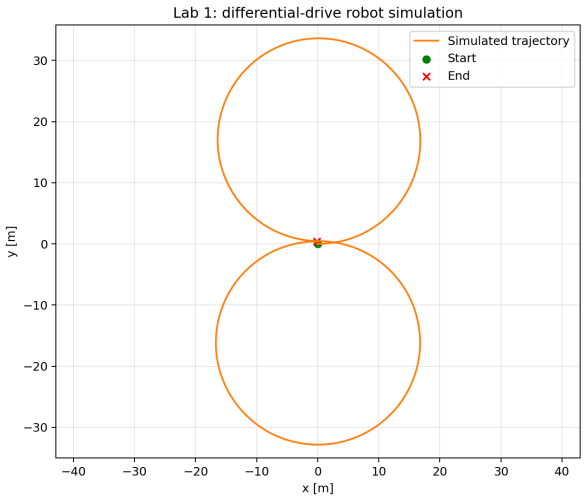



Figure 3: Trajectory generated directly by the simulated robot for the two opposite angular-velocity commands.

# 3 Lab 2: Dead-Reckoning Localization

## 3.1 Localisation sequence

The `Localization` base class defines the common iterative procedure. At every time step it:

1. advances the simulated robot through `robot.fs`;
2. obtains the encoder-derived input through `GetInput`;
3. updates the pose estimate through `Localize`; and
4. plots the estimated trajectory.

The derived `DR_3DOFDifferentialDrive` class supplies the two algorithm-specific methods in steps 2 and 3.

## 3.2 `GetInput`: encoder readings to body motion

The template documentation describes the input both as the raw encoder vector $[n_L, n_R]^T$ and as a generic four-component velocity. Neither is the value that `Localize` ultimately requires in this planar implementation. `GetInput` therefore reads $[n_L, n_R]^T$ from the simulated robot and converts it into the three-component body velocity $\boldsymbol{u}_k = \boldsymbol{\nu}_k = [u_k, 0, r_k]^T$.

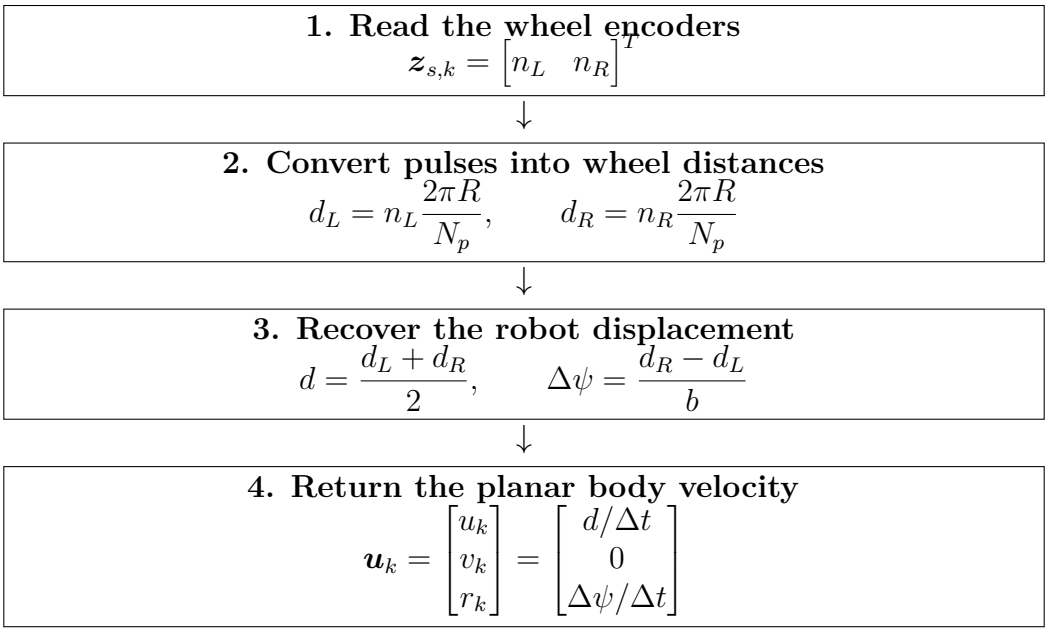



Figure 4: Conversion performed by `GetInput`.

The zero lateral component represents the non-holonomic constraint of the differential-drive platform. The implementation uses the same effective resolution, $N_p = 4096$ pulses per wheel revolution, to convert the measurements back into wheel distances. It also uses $R = 0.1$ m, $b = 0.5$ m, and $\Delta t = 0.1$ s.

## 3.3 Dead-reckoning pose update

The estimator integrates the encoder-derived velocity over one sampling period and compounds that body-frame displacement with the previous estimate:

$$\widehat{\boldsymbol{\eta}}_k = \widehat{\boldsymbol{\eta}}_{k-1} \oplus (\boldsymbol{\nu}_k \Delta t)\,. \tag{9}$$

The multiplication by $\Delta t$ converts velocity into a local pose increment. The $\oplus$ operation then rotates its translation according to the previous heading before updating the world-frame pose. The complete method is:

Listing 5: Complete `Localize` implementation.

```
def Localize(self, xk_1, uk):
    self.etak_1 = xk_1
    self.uk = uk
    nu_k = uk

    etak = Pose3D(xk_1).oplus(nu_k * self.dt)
    return etak
```

No compass, landmark or absolute-position measurement is used in this update. Consequently, every encoder error is integrated into the next pose and cannot be removed by the estimator.

# 4 Results and discussion

The comparisons use the same figure-eight command with 2094 steps per turn. Only the parameter indicated in each test is changed.

## 4.1 Effect of the velocity gain

$K$ is the feedback gain that corrects the difference between the current and commanded body velocities through $K(\boldsymbol{\nu}_d - \boldsymbol{\nu}_{s,k-1})$. It therefore controls how quickly the simulated robot responds to a new command. With $K = kI_3$, a value below one applies only part of the required correction, $k = 1$ applies it in one step, and a value above one overshoots the command before settling.

The test uses $k = 0.25$, 0.5, 1, and 1.5, while the encoder noise remains at its nominal value of 22 pulses. For each value, figure 5 shows the first turn and the complete figure-eight.

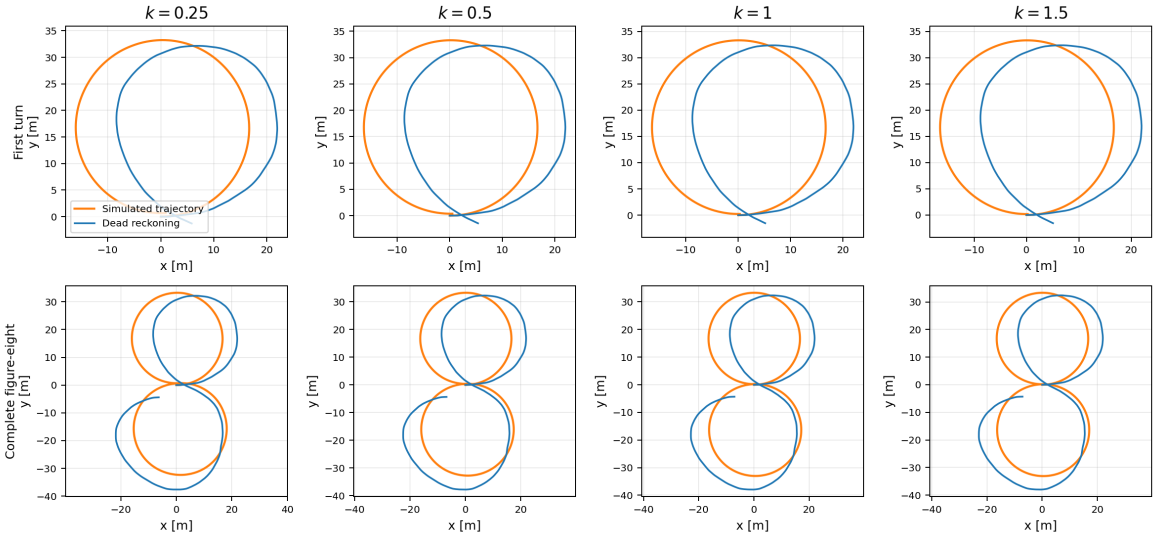



Figure 5: Simulated and dead-reckoning trajectories for the four tested velocity gains.

The tested gains produce similar trajectories because the velocity transient is short compared with the complete motion. Therefore, $K = I_3$ is retained as the simplest direct response.

## 4.2 Effect of the encoder noise

The second test fixes $K = I_3$ and repeats both trajectories with $\sigma_e = 0$, 11, 22, and 44 pulses. Here, $\sigma_e$ is the standard deviation of the additive encoder noise. The nominal value of 22 pulses comes from the default covariance $R_e = \mathrm{diag}(22^2, 22^2)$; it is a noise parameter rather than the encoder resolution of $N_p = 4096$ pulses per wheel revolution.

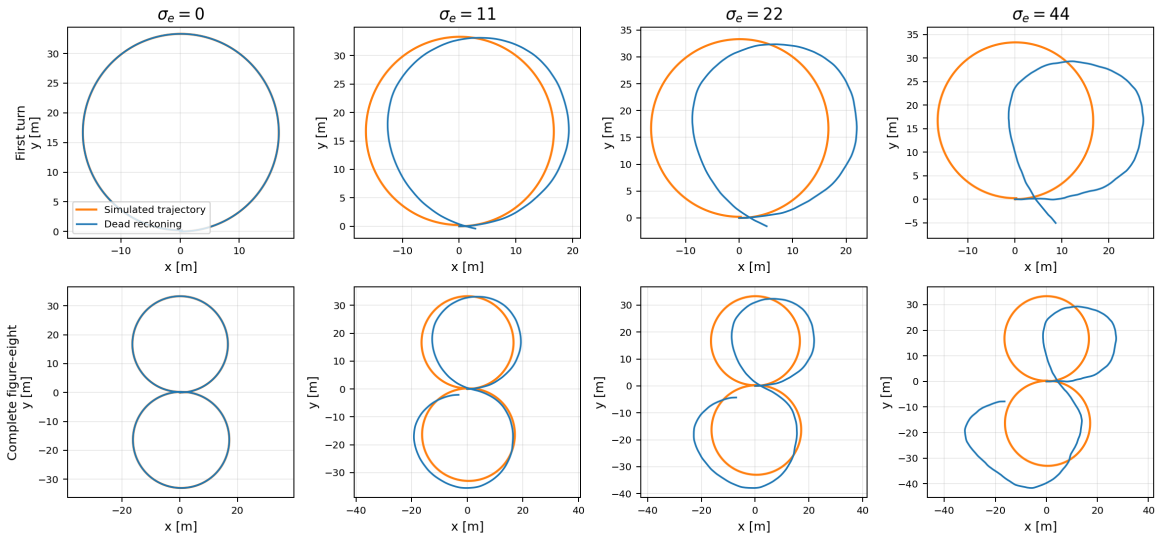



Figure 6: Simulated and dead-reckoning trajectories for the four tested encoder-noise levels.

With $\sigma_e = 0$, the simulated and estimated trajectories practically overlap. A small separation appears with 11 pulses, becomes clear at the nominal value of 22 pulses, and strongly deforms the estimate at 44 pulses. The smallest noise gives the closest result, while 22 pulses is retained as the nominal sensor setting. This comparison also shows that encoder noise has a much clearer influence on dead reckoning than the tested values of $K$.

# 5 Conclusions

The two laboratories establish a complete chain from planar rigid-body kinematics to simulated sensing and odometric localization. The `Pose3D` class provides consistent direct and inverse compounding, which is then reused by both the simulator and estimator. The differential-drive simulator generates noisy motion and encoder pulses, while the dead-reckoning class converts those pulses into incremental body motion and propagates the pose estimate. Circular and opposite-turn tests verify the expected robot motion and also demonstrate the fundamental limitation of dead reckoning: without an absolute observation, encoder noise produces unbounded drift.

# Resources

The source code used for these laboratories is publicly available in the Lab 1–2 GitHub directory:
https://github.com/Mariolopez31/PR-3501MO3352_2026/tree/main/Lab1-2